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Synthesis and Characterization of Zinc Oxide Nanostructured by Electrochemical Deposition Method

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Abstract. Zinc Oxide (ZnO) nanostructured was successfully synthesis by electrochemical deposition method. In this paper present the ZnO nanorods growth in a zinc nitrate/hexamethylenetetramine solution at 90°C with different potential applied. The effect of the growth process was investigated by the cyclic voltametric curve and the current-time curve. The structural of ZnO nanorods evidence that it has single crystalline, a wurtzite crystal structure with markedly preferential orientation along (001) direction was measured x-ray diffraction. The observation of ZnO nanorods was measured on field emission scanning electron microscopy that diameter of ZnO rods were below than 100 nm sizes. Uv-Vis spectrophotometer used to determine the transparency of ZnO nanorods through the UV light. The ZnO nanorods show the average transmittance (<90%) for all potential applied. The piezoelectric property of ZnO nanorods were measuring connected with two electrodes to the metal contact on the film that was driven by an ultrasonic wave. The piezoelectric output current was gained and characteristics curve have been illustrated for different voltage with constant driving frequency of ultrasonic wave at 40 kHz.

Introduction

Zinc oxide (ZnO) has become tremendously hot topic in recent years [1-4]. It is also well-know that one-dimensional ZnO nanowire arrays with larger length-to-diameter and surface-to-volume ratios than ZnO bulk and films have attraction due to their physical and optoelectrical properties. Many types of applications used ZnO nanostructure such as for laser [5], transistors [6], solar cell [7], nanogenerators and acoustics devices because of its strong piezoelectric effect. ZnO have been synthesis by a wide variety of technique such as pulsed laser deposition [8], sol-gel method[9], hydrothermal method [10] and electrochemical deposition method [11-14]. Among them electrochemical deposition method has advantages over other process because of its simplicity, the possible deposited in large area thin film and low cost equipment. In this present work, electrochemical technique was used to grown ZnO nanorods on the template that already deposited the ZnO by physical vapor deposition method. The structural, morphology, optical and piezoelectric properties have been studied using x-ray diffraction, field emission scanning electron microscopy, Uv-Vis spectrophotometer and ultrasonic wave measurement respectively.

Experiment details

Electrochemical deposition systems were consists with three electrodes. There is platinum rod as the counter electrode, glass substrate as the working electrode and silver/silver chloride (Ag/AgCl) was employed as the reference electrode. All these electrodes were connected with pententiostat an AUTOLAB (AUTO302N.FRA), monitored by the NOVA 5 software. Prior to the deposition, substrate were washed thoroughly with boiled, deionised water and rinsed before being cleaned ultrasonically. The aqueous solution contained 0.1M zinc nitrate, Zn $(NO_3)_2 \cdot 6H_2O$, HMT, potassium chloride (KCl) and dissolve 120 ml deionised water. The space width between the

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working electrode and counter electrode is about 2 cm, and effective area of the working electrode is 2 cm^2. At end of each growth period, the samples was removed from the solution and quickly rinsed in the following deionized water to eliminate any residual impurities from the surface.

The samples were prepared to study the influence of the deposition potential on the ZnO nanostructured films. The set of potential was deposited at -1.0, -1.1, -1.2, -1.3 and -1.4 V for 60 min, respectively. The structural of ZnO nanostructure was determined by x-ray diffraction Rigaku Ultima IV diffraction with source voltage /current (40kV/ 20mA) at scan in range 2θ from 20° to 80°. Morphology and sizes of deposit were studied by field emission scanning electron microscopy (FESEM) image, Supra 40VP. An optical property was investigated by using Uv-Vis spectrophotometer and piezoelectric property was examined by ultrasonic wave.

Results and discussion

Fig. 1 shows x-ray diffraction spectra patterns trend for applied potential at 2θ range from 20° to 80° by using 20mA/40kV source of CuKα type. The two diffraction peaks of (100) and (002) reveal at -1.0V and -1.1V, respectively. At -1.2, -1.3 and -1.4 only one peak of (002) reveal and (001) peak become low intensity. Good agreement of all potentials that corresponding of (JCPDS no 36-1451) data of the ZnO with hexagonal phase (a = 0.325 mm, c = 0.521 mm). The sharp and narrow (002) diffraction peaks at 34.22° are shown in every samples. This indicates that the synthesized ZnO materials are highly aligned perpendicular to the substrate with a c-axis growth direction.

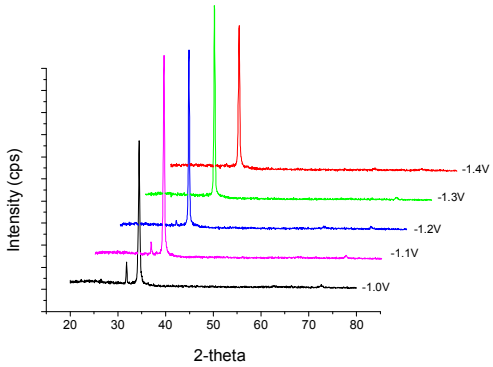

Fig 1. The XRD pattern for different potential applied

The morphologies of ZnO nanorods was display in (FESEM) images of the ZnO nanorods synthesized in different potential (a) -1.0V, (b) -1.1V, (c) -1.2V, (d) -1.3V and (e) 1.4V. The micrographs exhibited the uniform distribution of microstructures without any cracks and voids with the variation in rods size. The deposition of ZnO nanorods that can seen when more negative potentials the films was agglomerated as shown in fig. 2. From fig. 2(b), (c) and (d), that can see the deposition of potentials affected the grain size, more negative potentials applied to produce a larger crystalline grain and rougher film[15]. For all films has the grain size ranging approximate from 50 nm to 200 nm.

Optical transmittance curves of samples deposited at -1.0 V to -1.4 V are measured, respectively in Fig. 3. High optical transmittance of approximate 80% at 450 nm has been for film deposited at -1.0 V and -1.4 V. Transparency for thin film deposited at -1.3 V shows a rapid drop. Such a behavior is directly related to the structure of the film. This is maybe due to the noncompact nature of the film prepared at lower potentials, so the substrate area is not fully covered with ZnO and they transmit light.

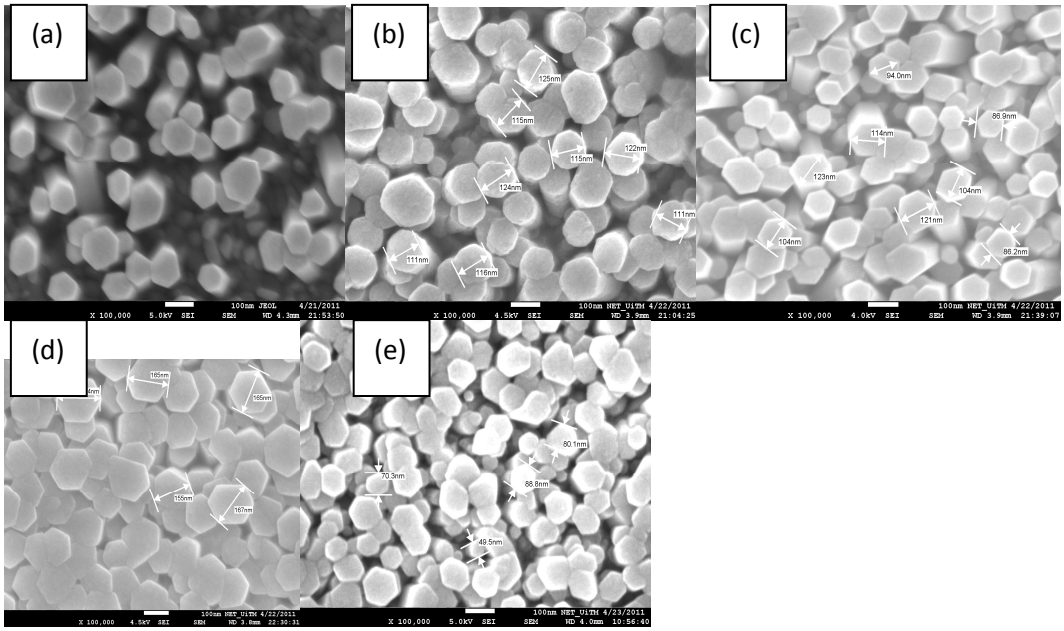

Fig. 2: FESEM images of ZnO nanorods (a)-1.0V, (b)-1.1V, (c) -1.2V, (d)-1.3V and (e) -1.4V. Growth condition are T= 90°C, t= 60 min

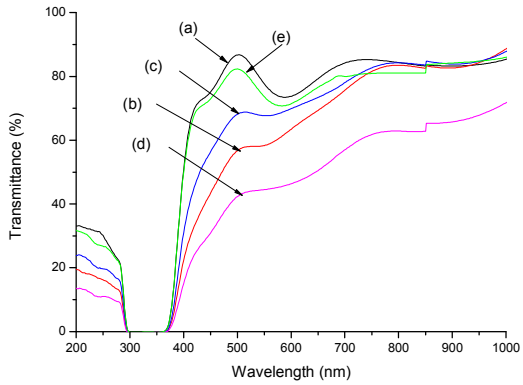

Fig. 3: Optical transmittance spectra of the electrodeposited ZnO films of various potentials (a) -1.0V, (b) -1.1V, (c) -1.2V, (d) -1.3 and (e) -1.4V

Fig.4 shows the short circuit current output measured for different potentials applied. The effect of the ZnO nanorods sizes and the surface-to-volume ratio on power generation of the piezoelectric property, the samples were by using ultrasonic wave. The ultrasonic wave frequency was fixed in 40 kHz in this experiment. When the ultrasonic wave has been turned off, the current of the device is estimated to be 0.05 pA due to perturbation of the measured environment, and it can be considered the background value. The current output is created when the ultrasonic wave is turned on, and the current disappears when the ultrasonic turned off. It can be seen that the average current output is 600 nA, 200 nA and 50 nA corresponding to potential voltage at -1.0, -1.2 and -1.1, respectively. At -1.3 and -1.4 V potential the output current is 0 means that films is weak when the current turn on.

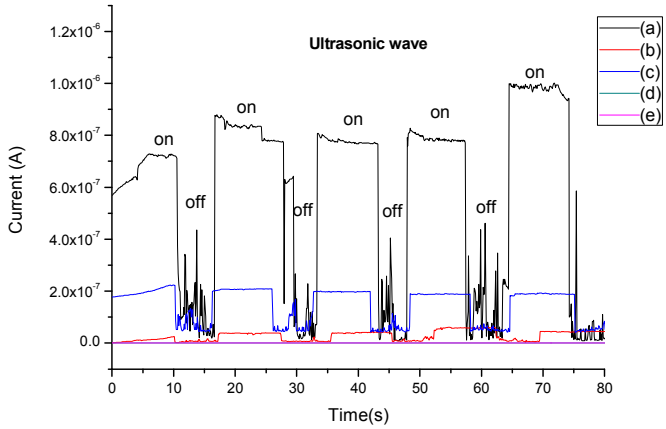

Fig4. Piezoelectric output current of different ZnO nanorods for different potential applied, (a) -1.0V, (b) -1.1V, (c) -1.2V, (d) -1.3V and (e) -1.4V

Summary

ZnO nanorods were successful deposited by electrochemical deposition method by different potential applied. X-ray diffraction show the ZnO nanorods has wurtizite hexagonal crystal structure because it growth at preferential orientation c-axis (002) plane. The morphology of ZnO nanorods can see the diameter was range about 50 nm to 150 nm for every potential. For optical properties was investigate by Uv-Vis spectra results was reveal that at -1.0 V and -1.4 V was good transparency in visible length. Piezoelectric properties was investigate by using ultrasonic wave that show at -1.0V potentials in average 600 nA current was measured on the ZnO films that effect the diameter and surface-to-volume ratio of ZnO that has an influence on the piezoelectric output efficiency.

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